

Introduction to DBMS & Data Models

1. What is DBMS?

A Database Management System (DBMS) is a software system that enables users to define, create, maintain, and control access to the database. It provides an environment that is both convenient and efficient to use. Examples include MySQL, Oracle, PostgreSQL, and SQL Server.

The primary goal of a DBMS is to provide a way to store and retrieve database information that is both convenient and efficient. By data, we mean known facts that can be recorded and that have implicit meaning.

2. Advantages over File System

Before DBMS, data was stored in file processing systems. This approach had several limitations:

- Data Redundancy and Inconsistency: Duplication of data in different files.
- Difficulty in Accessing Data: Need to write a new application program for each new task.
- Data Isolation: Data scattered in various files and formats.
- Integrity Problems: Difficult to enforce constraints.
- Atomicity of Updates: Hard to restore data to consistent state after failure.

3. DBMS Architecture

1-Tier Architecture

In 1-tier architecture, the DBMS is the only entity where the user directly sits on the DBMS and uses it. Any changes done here will directly be done on the database itself.

2-Tier Architecture

The 2-tier architecture is like a client-server application. The user interface program and application programs run on the client side, while the database system runs on the server side.

3-Tier Architecture

A 3-tier architecture separates its tiers from each other based on the complexity of the users and how they use the data present in the database. It includes a presentation tier (client), an application tier (business logic), and a database tier.

4. Data Models

Data models describe the structure of a database. Key types include:

- Hierarchical Model: Data is organized in a tree-like structure.
- Network Model: Similar to hierarchical, but allows a record to have more than one parent.
- Relational Model: Data is represented in two-dimensional tables (relations).
- Object-Oriented Model: Data is stored as objects.

5. Schemas and Data Independence

A database schema is the skeleton structure that represents the logical view of the entire database. Data independence is the ability to modify a schema definition in one level without affecting a schema definition in the next higher level.